

**SQL Interview Questions
For Data Analyst & Power BI
Developer
(1–3 Years Experience)**

SQL Basics

1. What is the difference between WHERE and HAVING?
2. What is the difference between DELETE, TRUNCATE and DROP?
3. What is NULL? How does it behave in comparisons?
4. What is the difference between COUNT(*) and COUNT(column)?
5. What is the difference between DISTINCT and GROUP BY?
6. What is a primary key?
7. What is a foreign key?
8. What is the difference between CHAR and VARCHAR?
9. What is the use of ORDER BY?
10. What is alias in SQL?

Joins

11. What is the difference between INNER JOIN and LEFT JOIN?
12. When do you use LEFT JOIN instead of INNER JOIN?
13. What is a SELF JOIN?
14. What is a CROSS JOIN?
15. What is CROSS JOIN?
16. How do you join more than two tables?
17. What happens if join columns contain NULLs?
18. Difference between JOIN and SUBQUERY
19. What is many-to-many relationship?
20. Why do we join fact and dimension tables?
21. How do you avoid duplicate rows after join?

Aggregation & Group By

22. What is GROUP BY?
23. Difference between WHERE and HAVING?
24. How do you find total sales by month?
25. How do you calculate average salary by department?
26. What is conditional aggregation?
27. How do you count distinct customers per city?
28. What is ROLLUP?
29. What is CUBE?
30. Difference between GROUPING SETS and UNION
31. How do you find Top 3 products by sales?

Subqueries & CTE

- 32. What is a subquery?
- 33. Difference between correlated and non-correlated subquery?
- 34. What is a CTE?
- 35. Difference between CTE and subquery?
- 36. Can we use multiple CTEs?
- 37. Can we update data using CTE?
- 38. When do you prefer CTE over joins?
- 39. What is a recursive CTE?
- 40. Can a CTE improve performance?
- 41. Can CTE be used in Power BI queries?

Window Functions

- 42. What is a window function?
- 43. Difference between ROW_NUMBER and RANK?
- 44. What is DENSE_RANK?
- 45. What is PARTITION BY?
- 46. What is LEAD and LAG?
- 47. How do you find running total?
- 48. How do you find highest salary per department?
- 49. Difference between GROUP BY and window function
- 50. What is OVER() clause?
- 51. When should you avoid window functions?

Date & Time

- 52. How do you get current date in SQL?
- 53. How do you extract month from a date?
- 54. How do you calculate last 30 days sales?
- 55. How do you calculate year-to-date sales?
- 56. How do you calculate previous month sales?
- 57. What is date difference?
- 58. How do you handle financial year (Apr–Mar / Jun–May)?
- 59. How do you calculate YoY growth?
- 60. How do you handle missing dates?
- 61. How do you convert string to date?

Data Cleaning

- 62. How do you remove duplicate records?
- 63. How do you replace NULL values?
- 64. What is TRIM()?
- 65. What is CASE WHEN?
- 66. How do you fix wrong data formats?
- 67. How do you split columns?
- 68. How do you merge columns?
- 69. How do you standardize city names?
- 70. How do you validate data in SQL?
- 71. How do you find bad data?

SQL for Power BI

- 72. Why should data be cleaned in SQL before Power BI?
- 73. What is query folding?
- 74. What are views?
- 75. Why create summary tables?
- 76. What is fact and dimension table?
- 77. What is star schema?
- 78. Why avoid many-to-many relationships?
- 79. How does SQL help in Power BI performance?
- 80. What is incremental load?
- 81. What is ETL?

Performance

- 82. What is an index?
- 83. How does index improve performance?
- 84. Why is a query slow?
- 85. What is an execution plan?
- 86. How do you optimize joins?
- 87. What is partitioning?
- 88. Difference between temp table and CTE?
- 89. What is table scan?
- 90. What is query timeout?
- 91. How do you handle large tables?